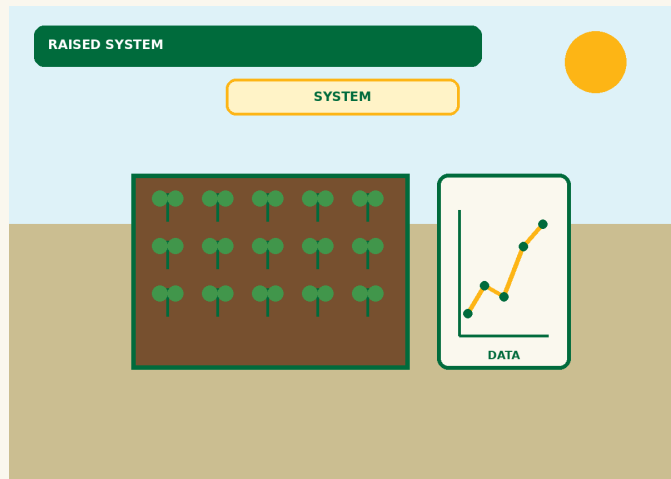




SYSTEM

RAISED SYSTEM

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 621 • RAISED SYSTEM

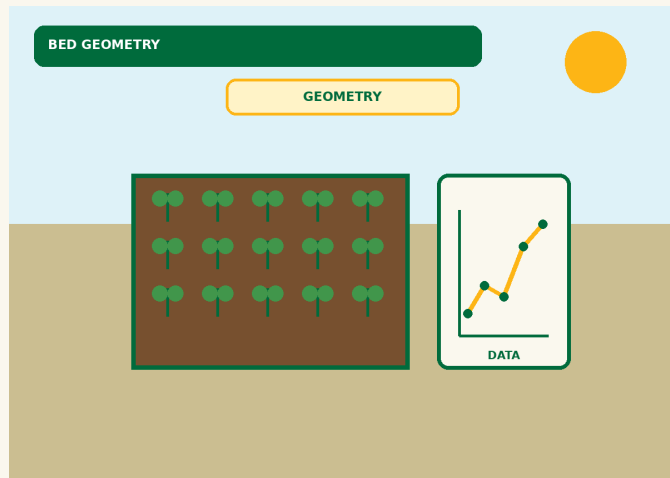
CONCEPT

A raised growing system is an engineered production space in which the root zone is elevated and physically defined.

INVESTIGATION TASK

Identify structural and biological components of the system.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



GEOMETRY

BED GEOMETRY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 622 • BED GEOMETRY

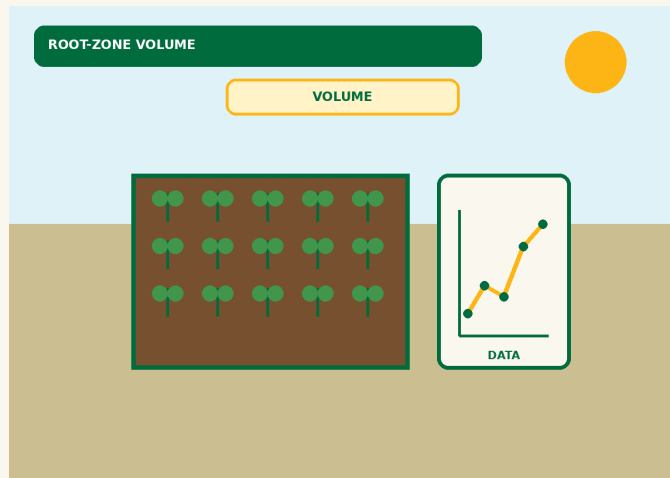
CONCEPT

Bed geometry determines surface area, soil volume, access, drainage pathways, and potential plant population.

INVESTIGATION TASK

Measure dimensions and calculate bed area.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



VOLUME

ROOT-ZONE VOLUME

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 623 • ROOT-ZONE VOLUME

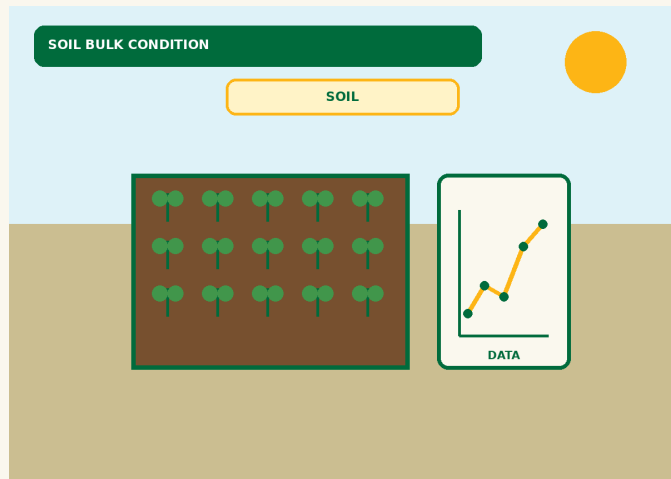
CONCEPT

Root-zone volume is the three-dimensional growing space available for roots and stored water, air, and nutrients.

INVESTIGATION TASK

Estimate soil volume from bed dimensions.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



SOIL

SOIL BULK CONDITION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 624 • SOIL BULK CONDITION

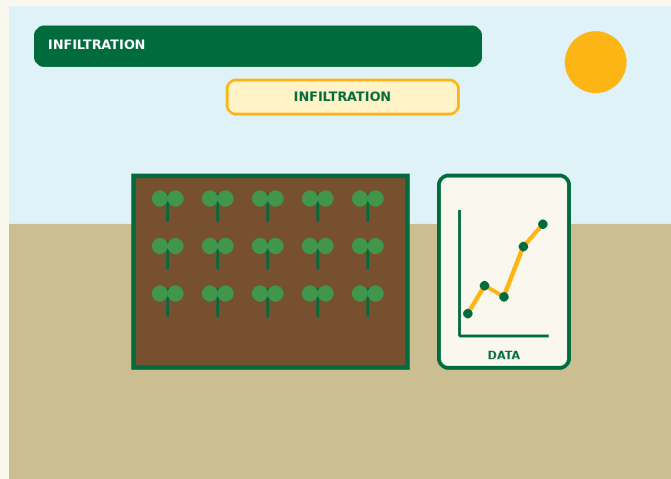
CONCEPT

Compaction and structure influence pore space, infiltration, aeration, rooting, and water storage.

INVESTIGATION TASK

Compare two bed locations using a safe field indicator.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



INFILTRATION

INFILTRATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 625 • INFILTRATION

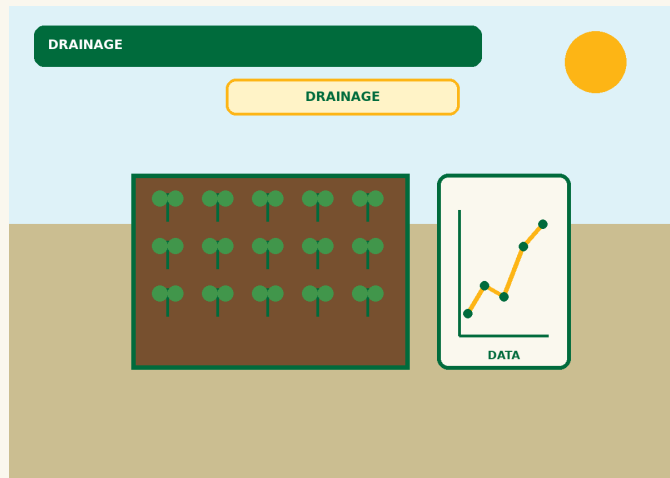
CONCEPT

Infiltration rate describes how quickly water enters the soil surface under specified conditions.

INVESTIGATION TASK

Conduct a teacher-supervised comparative infiltration test.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



DRAINAGE

DRAINAGE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 626 • DRAINAGE

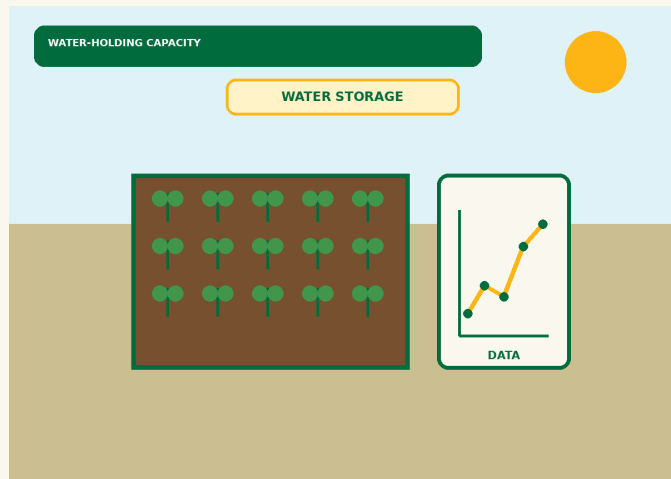
CONCEPT

Drainage removes excess water from the root zone and influences aeration and root health.

INVESTIGATION TASK

Trace the expected drainage pathway.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



WATER STORAGE

WATER-HOLDING CAPACITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 627 • WATER-HOLDING CAPACITY

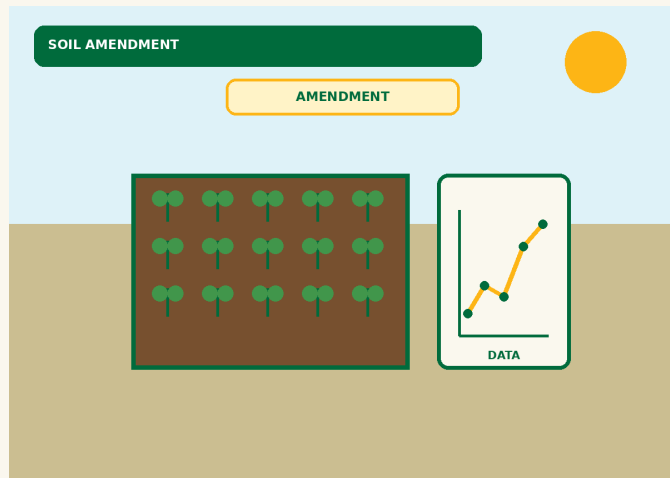
CONCEPT

Water-holding capacity describes the soil's ability to retain water after excess drainage.

INVESTIGATION TASK

Compare two growing media using a controlled demonstration.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



AMENDMENT

SOIL AMENDMENT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 628 • SOIL AMENDMENT

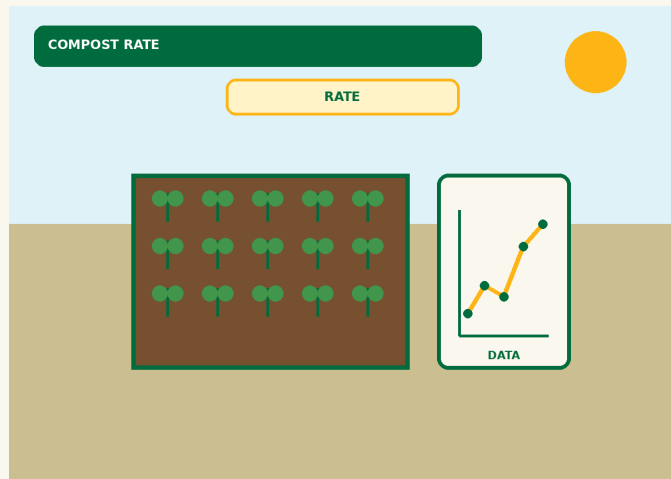
CONCEPT

An amendment is a material added to soil to modify physical, chemical, or biological properties.

INVESTIGATION TASK

State the intended function of one amendment.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RATE

COMPOST RATE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 629 • COMPOST RATE

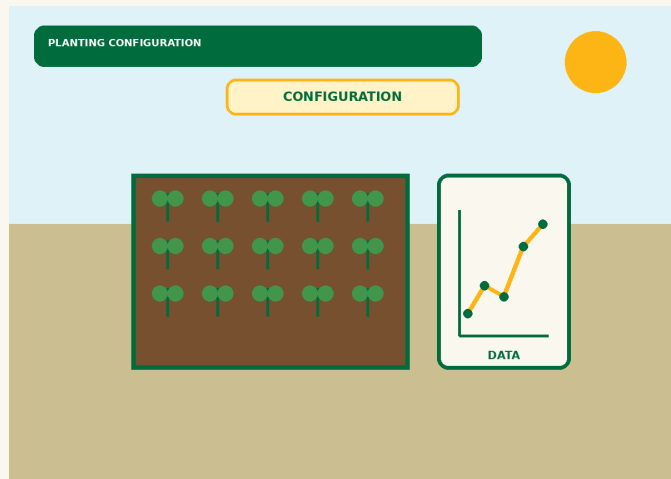
CONCEPT

Application rate expresses the amount of compost or amendment applied per unit area or bed.

INVESTIGATION TASK

Calculate a simple application rate.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CONFIGURATION

PLANTING CONFIGURATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 630 • PLANTING CONFIGURATION

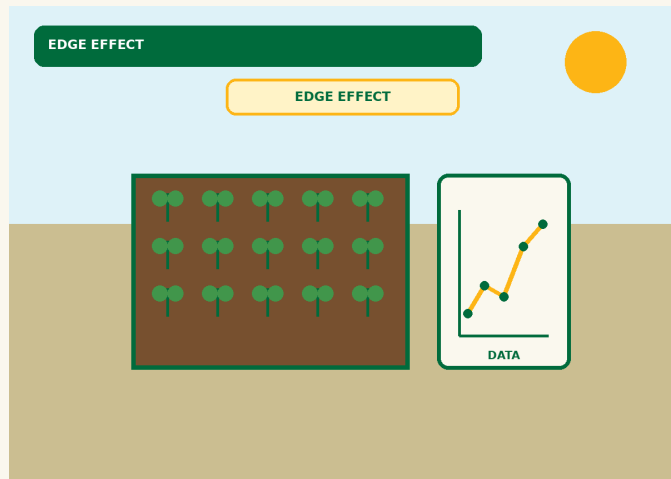
CONCEPT

Planting configuration combines row arrangement, spacing, crop architecture, and access requirements.

INVESTIGATION TASK

Compare two layouts for the same bed.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EDGE EFFECT

EDGE EFFECT

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 631 • EDGE EFFECT

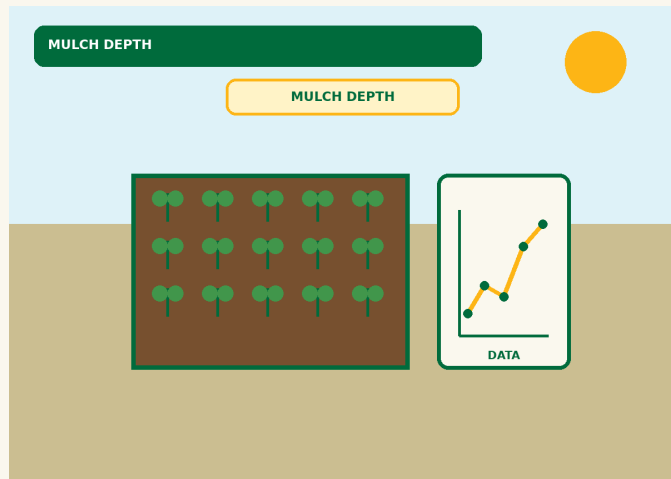
CONCEPT

Plants near bed boundaries may experience different moisture, temperature, light, or root conditions than interior plants.

INVESTIGATION TASK

Compare edge and interior observations.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MULCH DEPTH

MULCH DEPTH

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 632 • MULCH DEPTH

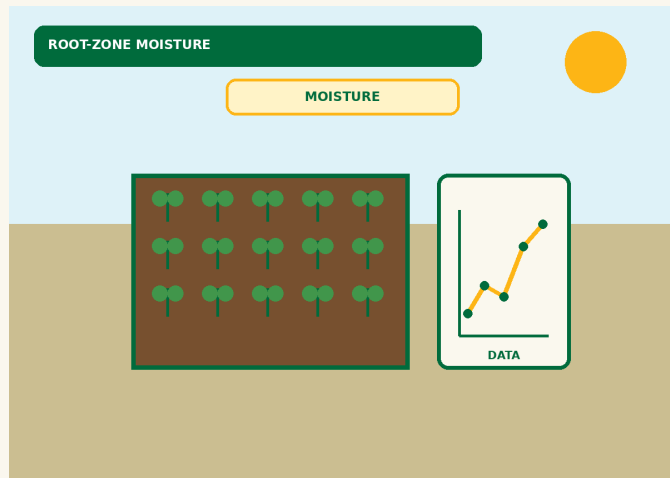
CONCEPT

Mulch depth influences soil cover, evaporation suppression, weed emergence, and water entry.

INVESTIGATION TASK

Measure mulch depth at several points.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MOISTURE

ROOT-ZONE MOISTURE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 633 • ROOT-ZONE MOISTURE

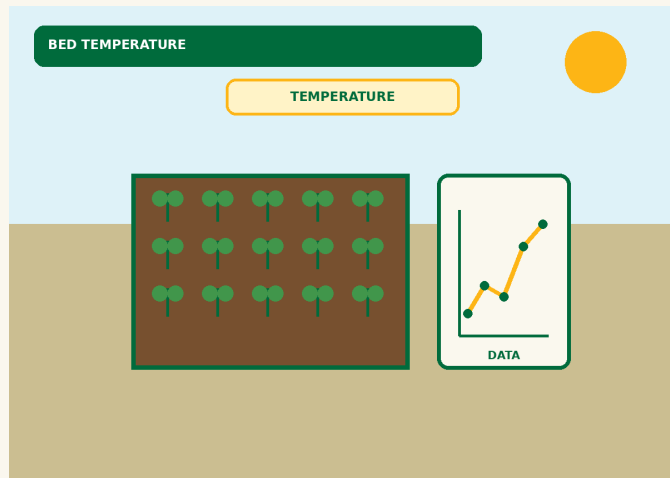
CONCEPT

Moisture can vary horizontally and vertically within a raised bed.

INVESTIGATION TASK

Compare readings or observations at multiple positions.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



TEMPERATURE

BED TEMPERATURE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 634 • BED TEMPERATURE

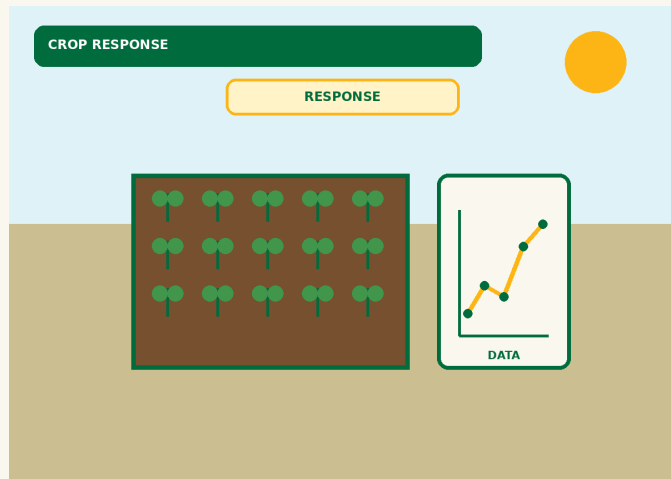
CONCEPT

Root-zone temperature can vary with depth, cover, water content, and exposure.

INVESTIGATION TASK

Compare teacher-approved temperature measurements.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RESPONSE

CROP RESPONSE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 635 • CROP RESPONSE

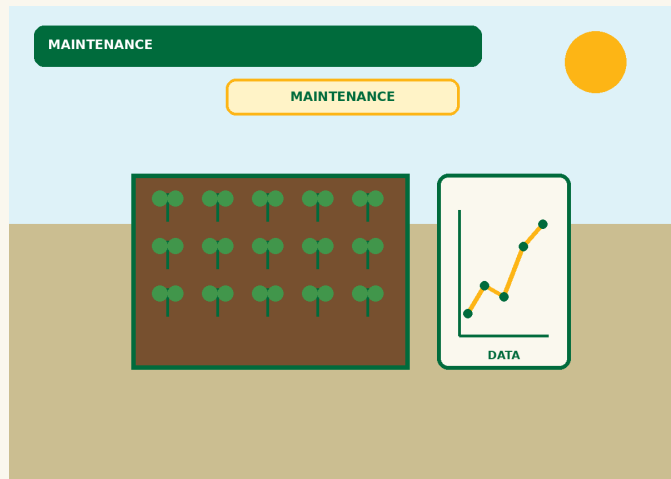
CONCEPT

Crop response is a measurable change in growth, condition, or yield associated with system conditions or management.

INVESTIGATION TASK

Relate one crop measurement to a bed condition.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



MAINTENANCE

MAINTENANCE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 636 • MAINTENANCE

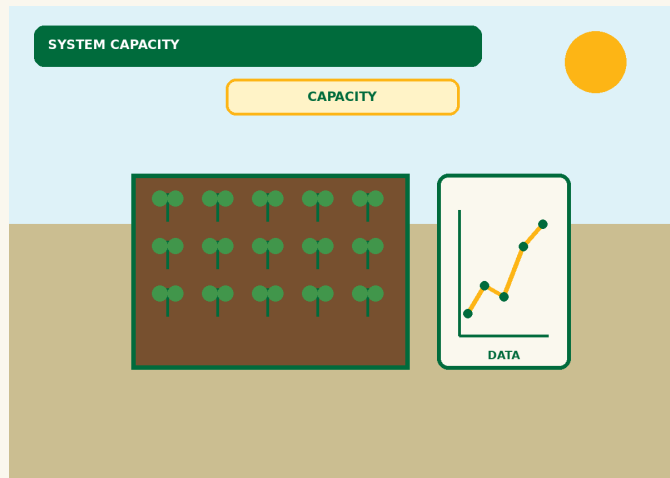
CONCEPT

Raised systems require structural, soil, irrigation, weed, sanitation, and access maintenance.

INVESTIGATION TASK

Create a maintenance checklist.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



CAPACITY

SYSTEM CAPACITY

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 637 • SYSTEM CAPACITY

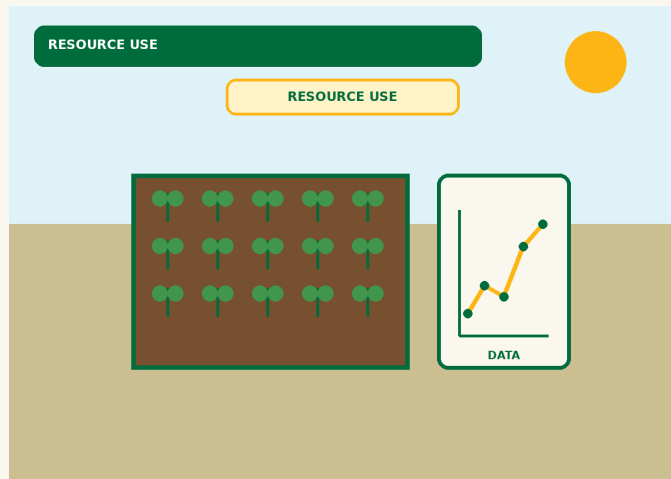
CONCEPT

System capacity describes how much growing area, root volume, plant population, or output the bed can support.

INVESTIGATION TASK

Calculate one capacity indicator.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



RESOURCE USE

RESOURCE USE

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 638 • RESOURCE USE

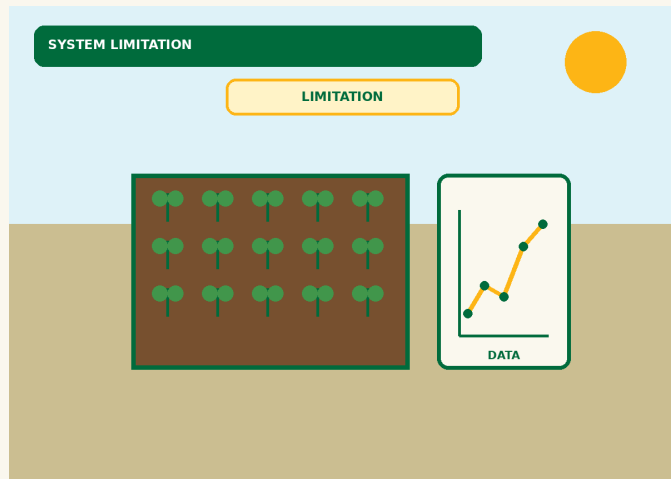
CONCEPT

Raised-system management can be evaluated by relating water, amendments, labor, or space to crop output.

INVESTIGATION TASK

Compare one resource-use metric between beds.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



LIMITATION

SYSTEM LIMITATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 639 • SYSTEM LIMITATION

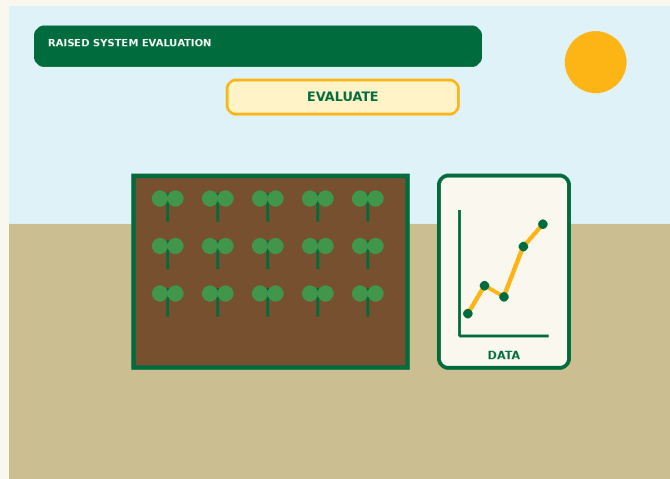
CONCEPT

A limitation is a structural, biological, or operational factor that constrains system performance.

INVESTIGATION TASK

Diagnose one observed limitation.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE



EVALUATE

RAISED SYSTEM EVALUATION

FORMULATE • TEST • MEASURE • ANALYZE • MANAGE

CARD 640 • RAISED SYSTEM EVALUATION

CONCEPT

Evaluation uses measured evidence to judge design, root-zone condition, crop response, resource use, and maintenance needs.

INVESTIGATION TASK

Recommend one design or management improvement.

QUESTION → MEASURE → RECORD → INTERPRET → DECIDE